A-A-50210A
18 May 1988
SUPERSEDING
A-A-50210
June 6, 1983

COMMERCIAL ITEM DESCRIPTION

ROD, LEVEL

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GG-R-1506.

This description covers Philadelphia-type level rods and targets with a separate marking strip which are components of survey equipment and are intended for use as self-reading or target rods for measuring vertical distances from a horizontal line of sight.

Salient characteristics:

Design: The level rods shall consist of two graduated sliding hardwood sections with clamping device to secure the sections in the fully extended position. The target shall be a micrometer type with vernier scale reading to 0.001 foot for English measurements and to 1 mm reading for metric measurements.

Styles:

I - English, graduations in feet, 10ths, and 100ths.
II - Metric, graduations in meters, decimeters, and centimeters.

Class:

A - With target
B - Without target

Part number. The document part number for the rod, level shall be as follows:

A50210 - X - Y

- Y — Class A or B.
- X — Style I or II.
- A50210 — CID identification number.

AMSC N/A FSC 6675

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.

Metric preferred because of deviation light

Style I level rod.

Graduations and numbers: All graduations shall be placed on the marking strip and shall be 0.01 foot wide and spaced 0.01 foot apart. The length and shape of the graduation marks shall vary so that the marks indicating 100ths, half 10ths, and 10ths of a foot can be readily distinguished from other graduation. The numbers shall read up on the front face of the rod from 0 to not less than 7 feet when the rod is closed and from 0 to not less than 13 feet when the rod is fully extended, except that the number "7" shall indicate the 13-foot graduation mark. Numbers on the rear face of the rod shall read down from 7 to 13 feet. Numbers indicating the feet shall be orange or red. All other numbers and graduation marks shall be black and the background shall be white. Typical markings are shown in figure 1.

Measurements: The rod shall have a 13-foot extended reading length and a 7-foot minimum reading length.

Style II level rod.

Graduation and numbers: All graduations shall be placed on the marking strip and shall be divided into meters, decimeters, and centimeters. Each meter shall be indicated by an orange or red numeral; decimeter numerals shall be black and the background shall be white. Typical markings are shown in figure 2.

Measurements: The level rod shall be 2 meters, plus or minus 0.2 meter, extending to 4 meters, plus or minus 0.3 meter.

Workmanship: All surfaces of the level rod shall be devoid of surface roughness and free from dents, sharp edges, or any other defect which could affect serviceability. Exterior corners and edges shall be rounded.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Packaging and marking. Military packaging shall be specified when shipments are to overseas Department of Defense (DoD) consignees. Commercial packaging may be specified for shipments to DoD consignees within the continental United States and to non-DoD consignees worldwide. When a supplier's normal packaging will satisfy the requirements for military packaging, it may be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.

Military packaging. Each level rod shall be preserved and packed to protect against physical damage and deterioration in a logistic environment characterized by the following:

1. Multiple handling during transportation and in-transit storage.
2. Shock, vibration, and static loading forces during worldwide shipment by all transport modes.
3. Uncontrolled warehouse storage for extended periods.
4. Weather exposure during shipment and in-transit transfers.
5. Stacking loads during shipment and storage.

Quantities of level rods of one style destined to a single addressee may be unitized into one or more unit loads.

Commercial. Each level rod shall be packaged in accordance with the supplier's normal method for retail distribution. Individually packaged level rods of one style shipped to a single destination may be unitized to permit handling as a single unit. Shipping containers shall comply with Uniform Freight Classification or National Motor Freight Classification rules.

Marking. Unit, intermediate (when provided), exterior packs shall be marked with the item nomenclature, stock or part number, quantity per pack, and contract number as a minimum, additional marking shall be as specified in the contract or purchase order. Exterior packs shall have in addition the complete consignee's address. Markings shall be applied in a legible manner, in a contrasting color, of an appropriate size, and of waterproof materials.

Quality assurance. Each Philadelphia-type level rod shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

1. Design not as specified.
2. Style not as specified.
3. Graduations and numbers not as specified.
4. Reading lengths not as specified under measurements.
5. Workmanship not as specified.
6. Material not as specified.
7. Packaging and marking not as specified.

Certification of commercial item. The contractor shall furnish certification for Government approval that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial marketplace for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.

Small business. A small business that cannot meet the certification-of-commercial-item requirement of this document, may make an offer under this solicitation; and, if such small business is the apparent successful offeror, the Government may require the small business to submit a sample level rod. Any such Philadelphia level rod may be evaluated for this solicitation if time to contract award permits. Such evaluation would be to determine conformance to all requirements of this commercial item description and to determine equivalency to Philadelphia rods available in the commercial marketplace in the areas of durability, human factors engineering, and safety. It should be noted that bid samples are not required unless specifically requested by the contracting officer. Note. The Government may elect not to evaluate a bid sample from an apparent low bidder for award for this solicitation if a determination of urgency is made by the contracting officer. The bid sample would then be evaluated prior to future solicitation for comparable items.

Notes:

1. When this commercial item description is used for procurement, the commercial item certification clause must appear in the solicitation.
2. Figures. The figures show the style of markings for Philadelphia-type level rods which have been found acceptable; however, the figures are for illustration only and are not intended to preclude the furnishing of other designs which conform to the salient characteristics.
3. Orderers should specify style and class required.

MILITARY INTERESTS:

Custodians

Army - ME
Navy - YD
Air Force - 99

Review Activities

DLA - GS
Navy - MC
Air Force - 82

CIVIL AGENCY COORDINATING ACTIVITY:

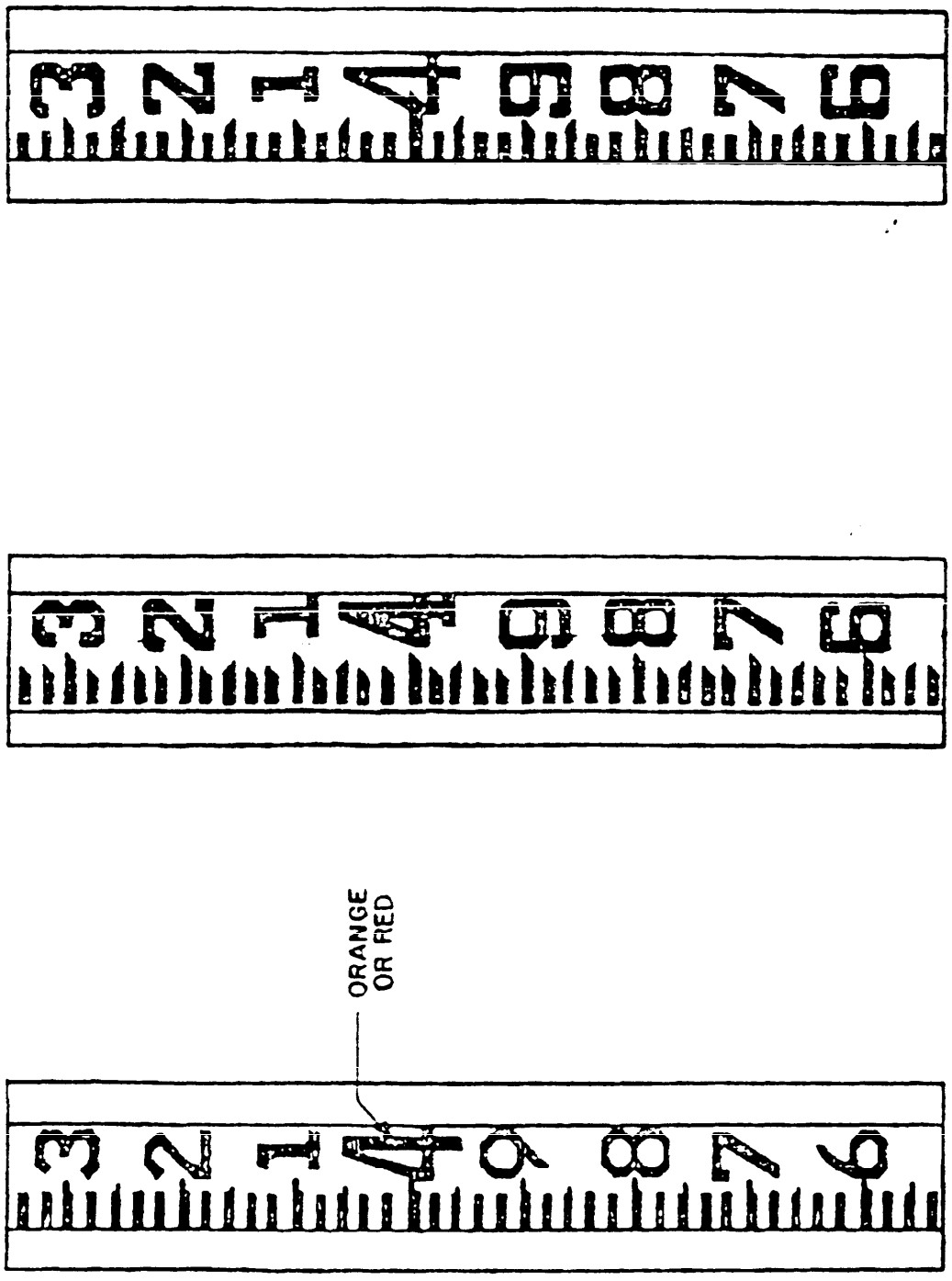
GSA - FQGC

PREPARING ACTIVITY:

Army - ME

Project 6675-0811

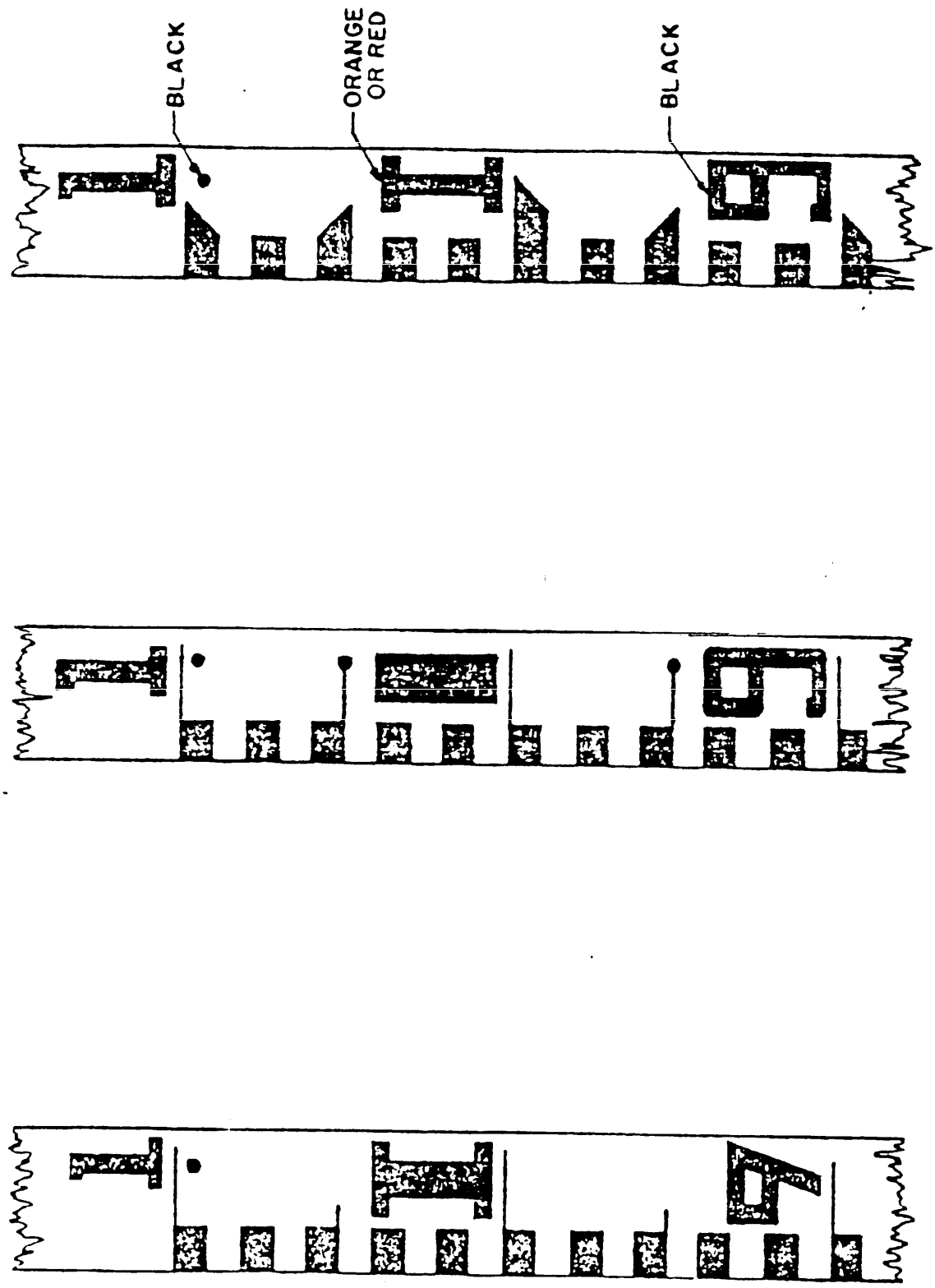
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.



X-32768

FIGURE 1. Typical markings for Style I Philadelphia level rod.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.



X-32778

FIGURE 2. Typical markings for Style II Philadelphia level rod.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:40Z
Check the source to verify that this is the current version before use.